

ENVARIABELSÄNALYS DEL 2

SEMINARIUM 26: funktioner av flera variabler

$$f: \mathbb{R}^n \rightarrow \mathbb{R}$$

EXEMPEL

$$a) z = \underbrace{x^2 + y^2}_{f(x,y)}$$

: Paraboloid

$$(f(x,y) = z, \mathbb{R}^2 \rightarrow \mathbb{R})$$

$$b) z = 1 + \sqrt{x^2 + y^2} : \text{kon} \quad (f(x,y) = z, \mathbb{R}^2 \rightarrow \mathbb{R})$$

$$c) z = \frac{x^2}{f(x,y)}$$

$$(\mathbb{R}^2 \rightarrow \mathbb{R})$$

" $\rightarrow$ " Tilldelning

$$d) z = \frac{1}{\underbrace{x^2 + y^2}_{f(x,y)}}$$

$$(f(x,y) = z, \mathbb{R}^2 \rightarrow \mathbb{R})$$

En graf till $z = f(x,y)$ är $T = \{(x,y,z) : (x,y) \in \mathbb{R}^2, z = f(x,y)\}$
 " : " sådana att

För att visualisera $z = f(x,y)$ använder man s.k. nivåkurvor till funktionen.

$$f(x,y) : \gamma_c = \{(x,y) : f(x,y) = c\}$$

Beteckning för nivåkurvor

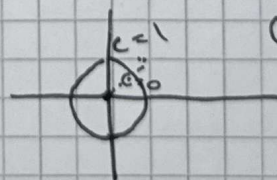
c är en konstant

EX: Skissa grafen till funktionen 1

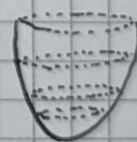
$$a) z = x^2 + y^2$$

$$\gamma_0 : x^2 + y^2 = 0 \leftarrow c$$

$$\gamma_1 : x^2 + y^2 = 1$$



OSV.



$$b) z = 1 + \sqrt{x^2 + y^2}, z \geq 1$$

$c=0$, $\gamma : f(x,y) = 0$ ger $1 + \sqrt{x^2 + y^2}$, händer aldrig.
 OBS $c \geq 1$

$$c=1 : \gamma_1 : 1 + \sqrt{0^2 + 0^2} = 1$$

OSV.



$$c=4 : f(x,y) = 4 \Rightarrow$$

$$1 + \sqrt{x^2 + y^2} = 4$$

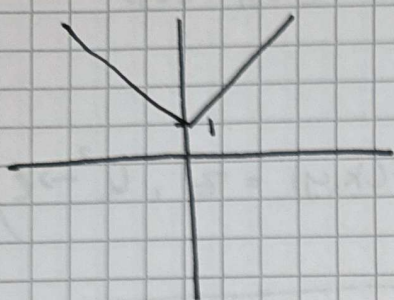
$$\Leftrightarrow \sqrt{x^2 + y^2} = 3 \quad c \Rightarrow$$

$$x^2 + y^2 = 3^2 \quad (\text{cirkel})$$

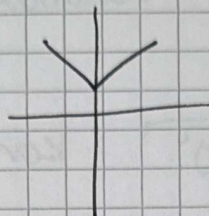
Oliva uyer ger da

$$x=0 \Rightarrow z = 1 + \sqrt{y^2} \Leftrightarrow z = 1 + |y| \Leftrightarrow$$

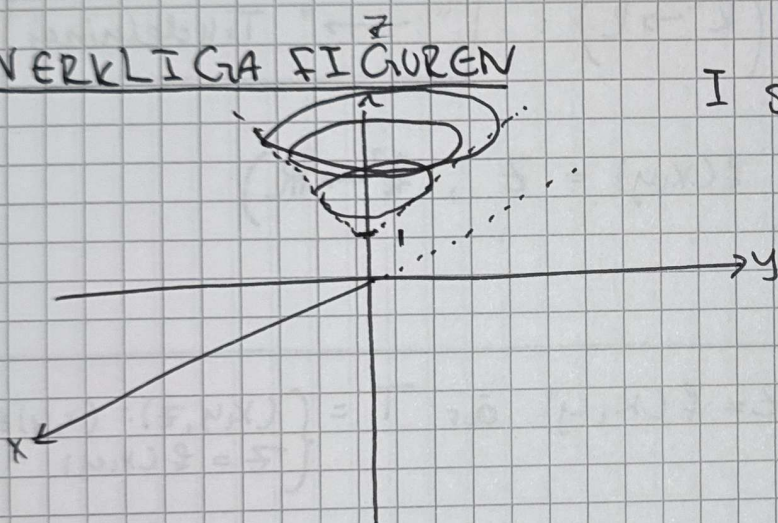
$$z = \begin{cases} 1+y & \text{om } y \geq 0 \\ 1-y & \text{om } y \leq 0 \end{cases}$$



$$\text{om } y=0 \Leftrightarrow z = 1 + |x|$$



VERKLIGA FIGUREN



I spetsen (0, 0, 1)

Partiella derivator,

$$y = f(x) \quad \text{der} \quad f'(x) = \frac{f(x)}{dx}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

FLERVARIABELANALYS

$$z = f(x, y)$$

$$f'_x = \frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \left(\frac{f(x+h, y) - f(x, y)}{h} \right)$$

$$f'_y = \frac{\partial f}{\partial y} = \lim_{h \rightarrow 0} \left(\frac{f(x, y+h) - f(x, y)}{h} \right)$$

DEF:

$f(x, y)$ är deriverbar om $f'_x(x_0, y_0), f'_y(x_0, y_0)$ existerar ändligt.

Andra derivata notation

$$\frac{\partial^2 f}{\partial x^2} \quad \text{eller} \quad \frac{\partial^2 f}{\partial x^2} \left(\frac{\partial f}{\partial x} \right)$$

Samma för y

$$f''_{xy} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right)$$

$$f''_{yx} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)$$